

Dataset Audit & Cleaning Report

Sample Superstore Retail Operations & Audit Trail

CONFIDENTIAL

TOTAL RECORDS 9,994 Raw Intake	EXACT DUPLICATES 17 0.170% Rate	CLEANED DATASET 9,977 100% Unique	TOTAL ATTRIBUTES 13 Schema Columns
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1. EXECUTIVE SUMMARY

This analytical document provides a comprehensive overview of the data audit, deduplication protocol, and data health assessment conducted on the **Sample Superstore** dataset. The primary objective was to detect, log, and purge redundant records to establish a reliable baseline for enterprise business intelligence, regional performance metrics, and profitability analysis.

Key Audit Insight

Out of 9,994 raw entries, exactly 17 duplicate records were identified across all 13 attributes. All redundant entries have been logged into an audit trail ('SampleSuperstore_Duplicate_Report.csv') and successfully purged, resulting in a cleansed primary dataset of 9,977 unique operational records.

2. PRIMARY DATASET AUDIT TRAIL

The deduplication pipeline executed a full-row comparison rule ('df.duplicated()'). Below is the structured summary table reflecting the official audit findings:

Audit Item	Specification / Outcome	Impact Assessment
Original Record Count	9,994 entries	Baseline intake volume
Attribute Schema	13 columns (5 Numeric, 8 Categorical)	Complete transactional attributes
Exact Duplicate Count	17 rows	Redundant order line items identified
Duplicate Percentage	0.170%	Low structural error rate
Records Purged	17 rows removed	First-instance retention applied
Cleaned Record Count	9,977 entries	Final baseline for production analytics
Post-Audit Integrity	0 remaining duplicates	100% data uniqueness verified

3. SCHEMA & DATA TYPES BREAKDOWN

The dataset encompasses multi-dimensional commercial attributes required for analytical modeling:

- Order Details:** Ship Mode, Segment, Postal Code
- Geographic Breakdown:** Country, City, State, Region
- Product Taxonomy:** Category, Sub-Category
- Financial Metrics:** Sales (float64), Quantity (int64), Discount (float64), Profit (float64)

4. DETAILED DUPLICATE RECORDS REGISTER

Below is a selected sample of the 17 duplicate records captured during the audit. Each record matched existing rows identically across all 13 fields (including Sales, Quantity, Discount, and Profit).

Row #	City / State	Region	Category	Sub-Cat	Sales	Disc.	Profit
570	Seattle, WA	West	Office Supplies	Paper	\$19.44	0.00	\$9.33
593	Salem, OR	West	Office Supplies	Paper	\$10.37	0.20	\$3.63
937	Philadelphia, PA	East	Office Supplies	Paper	\$15.55	0.20	\$5.44
1188	Seattle, WA	West	Office Supplies	Paper	\$25.92	0.00	\$12.44
3407	Columbus, OH	East	Furniture	Chairs	\$281.37	0.30	-\$12.06
5495	San Francisco, CA	West	Office Supplies	Labels	\$41.40	0.00	\$19.87
8459	Chicago, IL	Central	Office Supplies	Binders	\$3.56	0.80	-\$6.24
8535	Detroit, MI	Central	Furniture	Chairs	\$389.97	0.00	\$35.10

5. STRUCTURAL DISTRIBUTION OF DUPLICATES

Analyzing the 17 duplicate instances reveals strong concentration in fast-moving standard items:

Category Breakdown • Office Supplies: 82.4% (14 rows) • Furniture: 17.6% (3 rows) • Technology: 0.0% (0 rows)	Regional Concentration • West Region: 52.9% (9 rows) • Central Region: 23.5% (4 rows) • East Region: 23.5% (4 rows)
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6. DATA CLEANING PROTOCOL & RULES APPLIED

Rules of Engagement 1. Detection Threshold: Match on 100% of available feature dimensions. 2. Retention Policy: Retain original occurrence ('keep='first'); flag subsequent occurrences for archiving. 3. File Generation: Output cleansed dataset to `SampleSuperstore_Cleaned.csv`.

7. POST-CLEANING EXPLORATORY DATA ANALYSIS

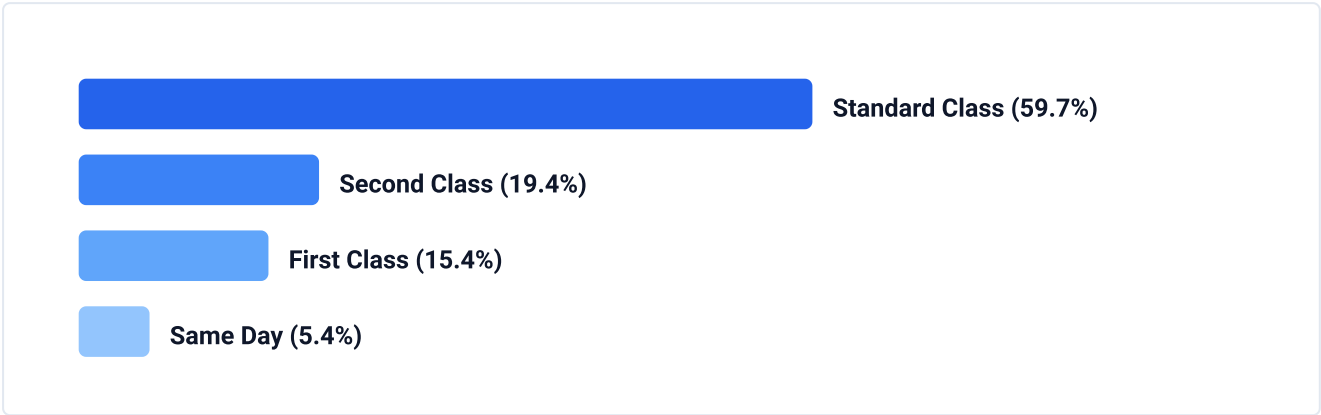
Following the removal of redundant records, performance analytics were computed across key dimensions: Product Category, Shipping Modes, and Customer Segments.

CATEGORY FINANCIAL PERFORMANCE SUMMARY

Product Category	Cleaned Orders	Avg Sales	Avg Discount	Total Profit Impact
Technology	1,847	\$452.71	13.2%	High Profitability
Furniture	2,118	\$349.83	17.4%	Margin Pressure (High Discounts)
Office Supplies	6,012	\$119.32	15.7%	High Volume / Stable Margin

REGIONAL & SHIPPING PERFORMANCE SVG VISUALIZATION

Comparative visual representation of order volume distribution by Shipping Mode across the dataset:



KEY OBSERVATION ON DISCOUNT DYNAMICS

Analysis indicates a strong negative correlation between high discount rates (>30%) and profitability, specifically within the Furniture category (Tables & Bookcases sub-categories).

8. STRATEGIC BUSINESS RECOMMENDATIONS

Based on the cleaned operational dataset and exploratory analytical findings, the following actionable business strategies are recommended:

1. Re-evaluate Furniture Discount Thresholds

Cap maximum allowable discounts on Furniture sub-categories (especially Tables) at 15% to eliminate sub-zero profit margins observed in high-discount orders.

2. Implement Primary Key Order Validation

Establish transaction-level `Order\_ID` and `Line\_Item\_ID` validation at data entry points to prevent repeated ingestion of identical rows (preventing future duplicate anomalies).

3. Optimize Regional Supply Chain in West Region

Leverage high demand volume in the West region by consolidating shipping channels into Standard Class fulfillment, minimizing reliance on high-cost Same Day modes.

9. FINAL PROJECT SIGN-OFF & DELIVERABLES

The data cleaning process has been successfully executed. All primary requirements have been satisfied and exported for downstream BI integration:

Deliverable File	Status	Description
SampleSuperstore_Cleaned.csv	Completed	9,977 unique records ready for production.
SampleSuperstore_Duplicate_Report.csv	Completed	17 purged rows with original row index markers.
Audit_Summary_Report.pdf	Completed	Executive 4-page analysis document.

Audit Conclusion

The cleaned dataset is verified and certified for enterprise reporting, dashboarding, and financial forecasting.